About Caterpillar®

Since 1925, Caterpillar Inc. has been helping our customers build a better world – making sustainable progress possible and driving positive change on every continent.

With 2019 sales and revenues of $53.8 billion, Caterpillar is the world's leading manufacturer of construction and mining equipment, diesel and natural gas engines, industrial gas turbines and diesel-electric locomotives. Services offered throughout the product life cycle, cutting-edge technology and decades of product expertise set Caterpillar apart, providing exceptional value to help our customers succeed. The company principally operates through three primary segments – Construction Industries, Resource Industries and Energy & Transportation – and provides financing and related services through its Financial Products segment.

Participant Evaluation Project

Problem Statement Title: Optimal Dump Packing

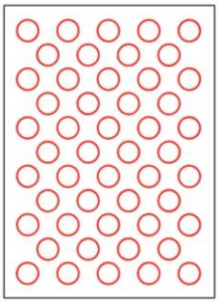
Problem Statement Description: Dump packing is a critical but highly manual step in mine planning today. While current tools provide basic packing functionality, they place a high cognitive and operational burden on users, especially when working with complex dump shapes. Current methods are based on a two step approach where

1) The dump is first filled with pre-defined "Spot Points"
2) Lanes are pre-planned to each Spot Point.

This approach has a number of limitations:
When small trucks dump to the pre-defined spot points, low spots occur within the dump. It may be impossible to pre-plan arrival sequences for mixed fleets.

Where the shape of a paddock dump is irregular, there will always be areas that cannot be covered by pre-defined spot points. The pre-defined spot points must be adequately separated so that dirt piles from prior dumps are not detected as obstacles (e.g. trucks stop). Due to the above limitations, methods based on the placement of pre-defined spot points may never match the density of staffed operations.

Alternative methods for filling dumps with spot points:
Staggered Pattern



Radial Fill Pattern

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Solution Options:
Option #1 – Propose a better method for packing spot points into polygons to achieve a packing density near to staffed operations. The solution does not need to support mixed fleets (e.g. may use a standard size for spot points). Lanes may be pre-planned for the defined spot points.
Option #2 – Propose an alternative method for creating a dump plan (not spot points) that allows Trucks in a mixed Fleet to achieve a packing density near to staffed operations. The design should provide guidance to Trucks regarding the sequence in which dumps are to be filled without defining specific spot points. Trucks in a mixed fleet will achieve a consistent packing density by dynamically setting spot points and planning lanes according to the dump plan.
Option #3 – No dump plan is defined. Trucks in a mixed fleet use a map (dynamically updated) or otherwise use their perception systems to set spot points and dynamically plan lanes to achieve a packing density near to staffed operations. Trucks may work independently or collaborate through peer-to-per communications.
All proposed solutions must satisfy the following requirements:

- The operational limitations of the Trucks must be taken into consideration (turning radius, …).
- Multiple Trucks must be able to dump concurrently given sufficient space (space not already covered by dumps (piles)).
  Trucks must not hit each other when working concurrently in a dump.
  Trucks must not deadlock when working concurrently in a dump.
- Trucks must dump without any part of the truck extending outside of the defined polygon.
- Trucks must not dump material in locations where the material (pile) may isolate sections of the polygon (block access to sections of the polygon).
- For options that support a mixed fleet, packing density must be maintained for arbitrary sequences of Trucks with different payloads.
- Truck perception systems may be enhanced to allow Trucks to dump arbitrarily close to prior dumps (piles). Truck systems may be trained to recognise piles (semantic segmentation) so that they may act differently when approaching piles.

Autonomous Dump Pattern with missed Dumps

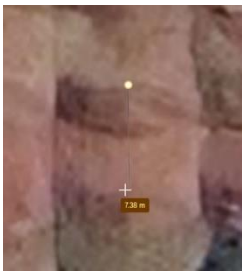


Staffed Dump Pattern with no missed Dumps



Autonomous Dump Spacing (7.38m)

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Staffed Dump Spacing (3.03m)



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